

CORE CURRICULUM



Finance

Mihir A. Desai, Series Editor

READING + INTERACTIVE ILLUSTRATIONS

Time Value of Money

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This reading contains links to online interactive illustrations, denoted by the icon above. To access these exercises, you will need a broadband Internet connection. Verify that your browser meets the minimum technical requirements by visiting <http://hbsp.harvard.edu/tech-specs>.

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1 INTRODUCTION

This chapter introduces the economics and mathematics of the time value of money: the idea that money has earning potential, so the timing of a payment matters. It is better to receive a sum of money today than to receive the same sum tomorrow because it can be invested and earn interest. Given an interest rate, we can calculate the present value of a sum to be received in the future or, alternatively, the future value of a sum invested today.

The notion of interest-bearing loans dates back many thousands of years to ancient agricultural societies in which “money” was grain and a farmer might borrow a portion at planting time, repaying the “loan” with a larger amount of grain at harvest time. As gold and other forms of money became more common mediums for exchange, basic credit and investing practices became more common as well. A merchant or family with more than enough wealth to meet its own immediate needs could earn a return on it by lending money at interest. By the Middle Ages, European banking houses had amassed enormous wealth through active, continuous borrowing and lending, and banking had become a business. Banks accepted deposits on which they paid interest, while simultaneously lending the funds at higher rates of interest. The bank earned the difference between the rate it received and the rate it paid. In this way, banks financed not only agriculture, but nearly every type of large-scale endeavor associated with human civilization, including trade, building, arts, exploration, and wars, all enabled by loans based on the simple idea of the time value of money.

Today, compounding and discounting—the two types of calculations used to determine the future and present value of money, respectively—are central to many applications in finance, including, for example: (1) investment finance—the valuation of securities such as stocks, bonds, and options; (2) corporate finance—the evaluation of corporate projects and even entire businesses, such as analyses of mergers and acquisitions, private equity transactions, alliances, and so forth; (3) personal finance—analyses of consumer loans such as home mortgages, car loans, credit card debt, and savings and retirement plans; and (4) public finance—the economics of government-sponsored plans such as pensions, healthcare, and unemployment insurance.

Despite this wide variety of issues and applications, the fundamental concepts and mathematical tools are the same across them all. This reading presents simple examples drawn mostly from personal finance, but the ideas and formulas are applicable to all of the areas mentioned above.

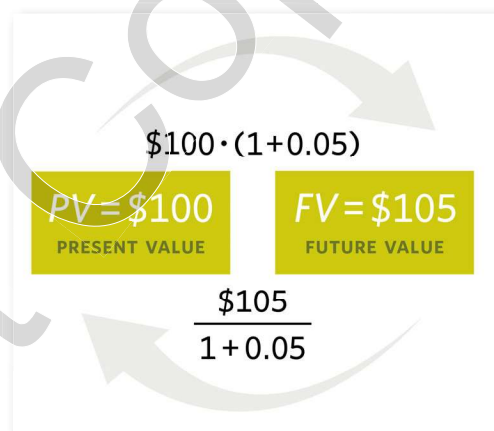
2 ESSENTIAL READING

2.1 Present and Future Value Over a Single Period

The notion that money has **time value** is a fundamental concept in finance. Someone who has \$1 may consume \$1 worth of goods today, or save and invest it for consumption at a later date. If the dollar is saved at the beginning of the period, it can earn interest at the **risk-free interest rate**, r , so its value at the end of the period will be greater than \$1. This extra value—the return on risk-free investing—is time value.

To illustrate, if we invest \$100 today at a risk-free rate of 5% per year, the value of our investment at the end of one year will be $\$100 + \$100 \cdot 0.05 = \$100 \cdot (1 + 0.05) = \105 . This \$105 is the **future value (FV)** of our investment, one period (one year) from now. Its value today (\$100) is its **present value (PV)**. (See Exhibit 1.)

EXHIBIT 1 The Relationship Between Present and Future Values Over a One-Year Period (assuming a risk-free rate of 5%)



More generally, for a single period:

$$FV = PV \cdot (1 + r)$$

where FV = future value, at the end of the period

PV = present value, at the beginning of the period

r = the risk-free rate of return per period

Conversely, if we are certain to receive \$105 one year from today and the risk-free rate is still 5%, then we can use the same relationship to compute its present value today. It must be

$\$105 / (1 + 0.05) = \100 . The value today of \$105 to be received one year from now must be less than \$105. In the first instance, we are moving from present to future value; in the second, we are going from future to present value. In both, we are invoking the simple relationship among PV , FV , and r , the time value of money over a single period:

$$FV = PV \cdot (1 + r)$$

or equivalently

$$PV = \frac{FV}{1 + r}$$

This equation reflects the fundamental meaning of the familiar saying: “A dollar today is worth more than a dollar tomorrow.” That is, if you receive the dollar today, you can invest it and you will have *more than a dollar* tomorrow. Future and present values are related to one another by their locations in time and the risk-free rate of interest over that period of time.

Let’s return for a moment to our earlier discussion of the risk-free rate, which we have defined as the rate on an investment that is free from default risk.^a Examples of risk-free investments include US Treasury securities, which many investors consider to be the closest thing to a default-risk-free instrument, and certain US government–insured bank deposits.

Throughout this reading the term *risk-free rate* refers to a **nominal rate of return**. The nominal rate of return expressly ignores potential changes in the purchasing power of the associated currency; that is, we may be certain to receive \$105 one year from now, but we are not guaranteed that the dollar will have the same purchasing power a year from now that it has today.^b We expect investors to care about real returns, so the distinction between real and nominal returns is an important one in the real world. However, the complication of risks involving currency values does not change the fundamental principle of time value or the mathematics of compounding and discounting. Accordingly, in this reading, we will work solely with nominal cash flows and nominal interest rates.

^a Risk is an important consideration in many financial analyses and will be covered in depth in other Core Readings. However, to focus first on the basic tools needed to understand the time value of money, risk is mostly ignored in this reading. To develop tools for discounting and compounding, we work with default-risk-free securities. [We note, though, that even default-risk-free securities may still be subject to other risks, such as fluctuations in a currency’s purchasing power: inflation.] Section 2.5 (Perpetuities and Annuities) presents examples in which risk is present and reflected in interest or discount rates, but does not otherwise affect the structure of the problem or the necessary calculations.

^b For this reason, the nominal rate is sometimes decomposed into a **real rate of return** and an **inflation rate**. The nominal risk-free rate equals a real risk-free rate plus an inflation rate: $r = r_r + i$, where r_r denotes the real (i.e., inflation-adjusted) return and i denotes inflation.

2.2 Compounding and Discounting Over Multiple Periods

If we invest \$100 for two periods instead of one, at the end of the second period we will have $[\$100 \cdot (1+r)] \cdot (1+r)$ or $\$100 \cdot (1+r)^2$. Not only will we earn interest on \$100 for two periods instead of one, but we will also earn *interest on interest* during the second period. If the risk-free interest rate is 5% for both periods, we will have $\$100 \cdot (1.05)^2 = \110.25 at the end of the second period. After a third period, we will have $\$110.25 \cdot 1.05 = \$100 \cdot (1.05)^3 = \$115.76$, and so forth. Earning interest on interest, known as **compounding**, reflects the accumulation of time value over multiple periods.

Interactive Illustration 1 computes the value of \$100 at various points in time—one, two, or three years into the future (the FV of \$100) or one, two, or three years in the past (the PV of \$100) at various rates of interest. At points in the future, the value is greater than the initial \$100 investment owing to the effects of compounding. At points in the past, the value is less than \$100 owing to the effects of **discounting**, the calculation of the (lesser) present value associated with a given (greater) future value. In discounting calculations the interest rate is often referred to as the **discount rate**.

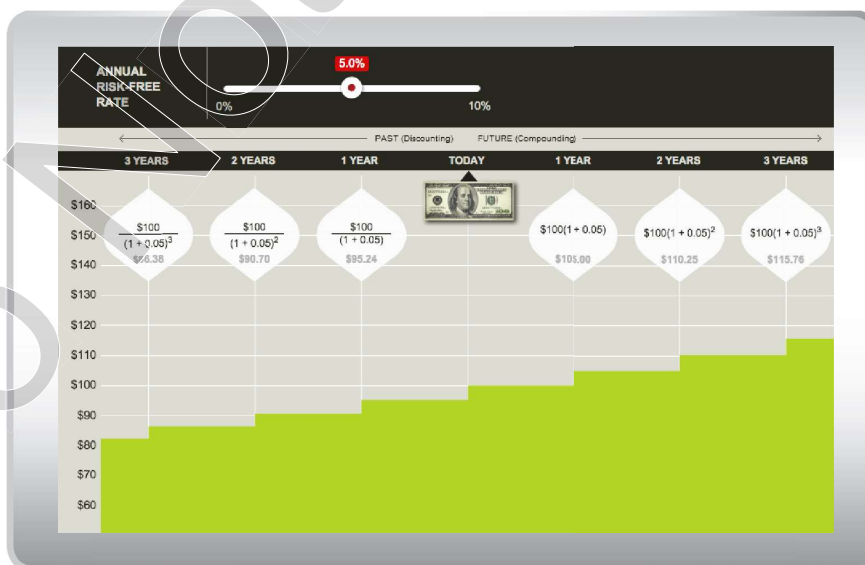
Interactive Illustration 1 starts with a risk-free rate of 5%. Experiment with the effects of discounting and compounding at different rates by moving the interest rate slider. What effects do discounting and compounding have on the value of \$100 at 0%, 5%, and 10%? When are the effects greater—at higher or lower interest rates?



INTERACTIVE ILLUSTRATION 1 Time Value of \$100 Cash



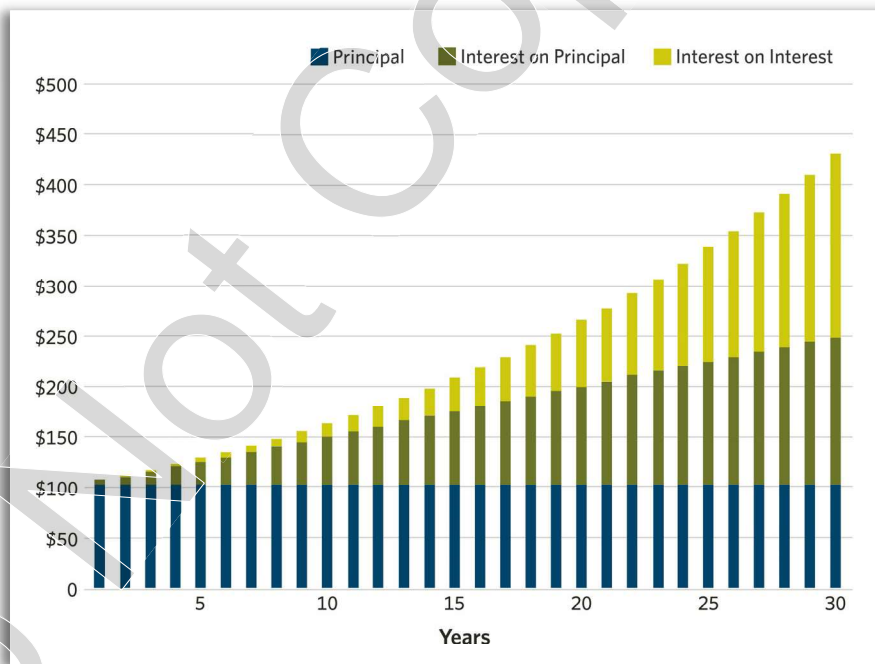
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The power of discounting or compounding over a long period is more substantial than many people realize. Suppose we invested \$100 for 30 years at 5% per year. At the end of 30 years, we would have $\$100 \cdot (1.05)^{30} = \432.19 . The initial investment grows by a factor of more than four, from \$100 to more than \$400. Where does this impressive increase come from? Note that 5% interest on \$100 is only \$5 per year, and 30 years of such interest is 30 times \$5, which equals \$150. But we actually earn *more than twice* that amount (we earn $\$432.19 - \$100 = \$332.19$) because of compounding: interest on interest.

Exhibit 2 decomposes the growth of \$100 invested at 5% per year for 30 years, showing how much of the total value each year comprises: (1) **principal**—the original \$100, depicted by the blue bars; (2) interest on principal, represented by the green bars; and (3) interest on interest, shown by the yellow bars. The blue bars remain constant at \$100. The green bars grow by \$5 each year—after 30 years, the green bar has reached \$150. But the yellow bars—interest on interest—grow the fastest. The yellow bar starts at zero, but by year 30 it is larger than either the blue or the green bars. The longer we continue to invest, the greater the power of compounding. And at higher rates of interest, the power of compounding develops even earlier.

EXHIBIT 2 The Power of Compounding



To generalize, over T periods of time, the future value of a risk-free investment is computed by multiplying the original investment (the PV) by a **compounding factor** equal to $(1+r)^T$:

$$FV = PV \cdot (1+r)^T$$

Conversely, we can rearrange this formula to calculate the present value of a sum to be received T years in the future as

$$PV = \frac{FV}{(1+r)^T}$$

For example, if we know we will receive \$1,000 30 years from now and the risk-free rate is 5%, we can calculate its present value as

$$PV = \frac{FV}{(1+r)^T} = \frac{\$1,000}{(1.05)^{30}} = \$231.38$$

In this calculation, we multiplied the future value by a **discount factor** equal to $1/(1+r)^T$ to calculate the present value today of an amount to be received T periods in the future:

$$\text{Discount factor} = \frac{1}{(1.05)^{30}} = 0.23138$$

Multiplying \$1,000 by 0.23138 gives a present value of \$231.38.

Now consider a higher risk-free rate, 7.5%, applied to the same \$1,000 over $T = 30$ years. The effect on the PV is substantial:

$$PV = \$1,000 \cdot \frac{1}{(1.075)^{30}} = \$1,000 \cdot 0.11422 = \$114.22$$

The present value is less than half as large at a discount rate of 7.5% as compared to 5%. In this example, the discount factor went from $1/(1.05)^{30} = 0.23138$ to $1/(1.075)^{30} = 0.11422$ and the PV accordingly fell by more than half, again showing the power of discounting over an extended period.

2.3 Multiple Cash Flows and Multiple Periods

The same principle of time value and tools for compounding and discounting apply to multiple cash flows. We simply apply the appropriate discount or compounding factor to each cash flow, according to its location in time, and then sum the results.

Suppose you wish to buy a house five years from now, and you need to save money for a down payment of \$40,000. You initiate a program today in which you save the same amount each year and invest it at the risk-free rate of 5%. How much do you need to save each year to accumulate \$40,000 by the end of year 5? Clearly, \$8,000 would be more than enough—five

years of saving \$8,000 per year will amount to \$40,000 even without interest. But will \$7,000 per year be enough? Let's set up the problem in a diagram and a table to find out. Let:

C_f = cash flow saved and invested at the beginning of each year = \$7,000

t = yearly points in time, starting today (i.e., t takes values 0 through 5)

T = total number of periods = 5

$r = 5\%$

It is important to be clear about whether each cash flow occurs at the beginning or end of the period. In this example, we put aside \$7,000 now ($t = 0$) and at the beginning of each subsequent year. That means we invest the first \$7,000 for five years, the second \$7,000 for four years, and so on, with the last \$7,000 invested for one year.

EXHIBIT 3 Sample Savings Plan to Buy a House in Five Years

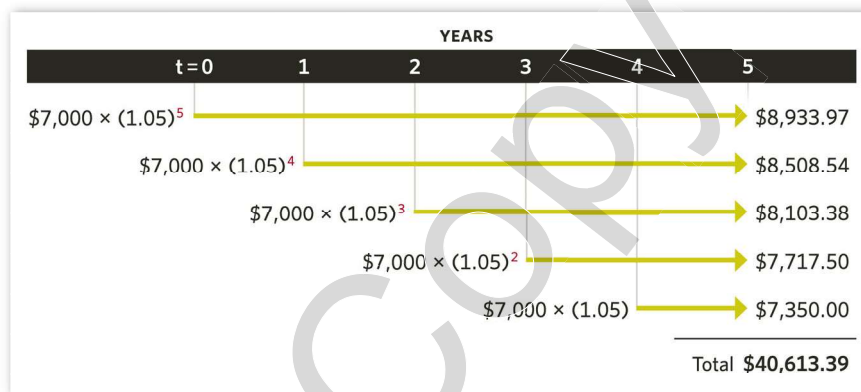


Exhibit 3 uses timelines to depict the save-and-invest scheme. **Exhibit 4** lays out the relevant figures for each year.

EXHIBIT 4 Calculating the Future Value of a Five-Year Savings Plan

Point in Time (yearly)	New Investment During the Year	Number of Years in Which Interest Is Earned	Compounding (FV) Factor	Future Value at the End of the Program
t	C_f	$T - t$	$(1 + r)^{T-t}$	FV
0 (today)	\$7,000	5	1.2763	\$8,933.97
1	\$7,000	4	1.2155	\$8,508.54
2	\$7,000	3	1.1576	\$8,103.38
3	\$7,000	2	1.1025	\$7,717.50
4	\$7,000	1	1.0500	\$7,350.00
5 (total)				\$40,613.39

Exhibit 4 shows that the future value of our save-and-invest scheme will be \$40,613 at the end of five years—more than enough to make the desired down payment. The table shows how long each year’s savings of \$7,000 will be invested at a rate $r = 5\%$ from the time it enters the pool until $t = 5$. Each year’s \$7,000 of savings is then multiplied by the appropriate compounding factor. Summing the results gives the future value of \$40,613.39.

For multiple periods and multiple cash flows, the accumulated future value at time T of a series of periodic cash flows is given by:

$$FV_T = Cf_0(1+r)^T + Cf_1(1+r)^{T-1} + \dots + Cf_{T-1}(1+r) = \sum_{t=0}^{T-1} Cf_t(1+r)^{T-t}$$

where T = the total number of periods (the endpoint)
and t = the date at which a cash flow occurs.

If we put aside the first \$7,000 at the end of the first year instead of at the beginning, our entire save-and-invest plan would be shifted out one year—in effect, it would be a four- rather than a five-year plan—and would give a different result (\$31,679 instead of \$40,613 because we would lose the \$8,933.97 associated with investing \$7,000 from $t = 0$ to $t = 5$). This would not meet our goal for the down payment. Not surprisingly, in time value calculations, the timing of cash flows is important.

We could use the formula, a table like Exhibit 4, or spreadsheet software such as Excel to answer “what if” questions, such as, What if the risk-free rate was only 3%? In that case we would need to save more: \$7,315 per year instead of \$7,000. Or, What if we needed a down payment of \$50,000 instead of \$40,000? At a risk-free rate of 5%, we need to save much more: \$8,618 per year.

Let’s now compute present values instead of future values. What if you are offered a security that will pay you \$100 at the end of each of the next five years, for a total of \$500? Such a security is called an **annuity**—a constant payment each period for a finite number of periods. How much should you be willing to pay for it? Clearly it is worth less than \$500 because of the time value of money—you won’t receive all \$500 today, so it cannot be worth \$500 today.

Interactive Illustration 2 demonstrates the discounting calculations required to compute the present value of a security that pays \$100 per year for five years. When you click “Run,” the illustration shows how each year’s \$100 cash flow is multiplied by the appropriate discount factor to obtain its present value; the present values of each are summed to give the PV for the entire security (\$432.95 at an interest rate of 5%). Now change the risk-free rate and click “Run”

again. Higher rates result in lower present values, and vice versa, but the formulas remain the same.

Note that the discount factor (or **PV factor**) used in Interactive Illustration 2, $1 / (1+r)^t$, has a value less than 1.0 for each time period. Further, this factor decreases as the rate or the length of the period increases.

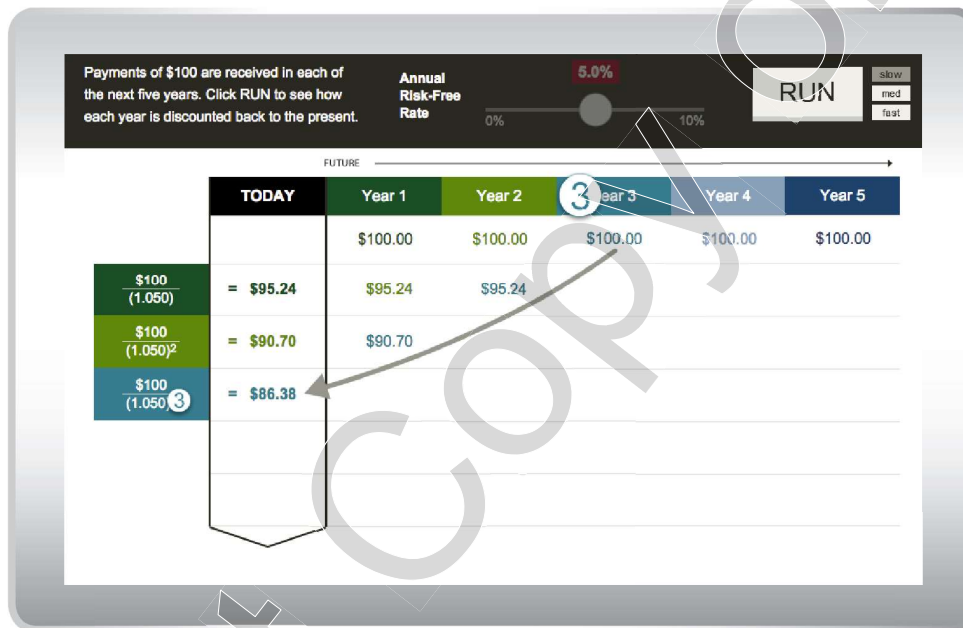


INTERACTIVE ILLUSTRATION 2

Calculating the Present Value of Multiple Cash Flows



Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2pKuSIo



Interactive Illustration 3 lets you choose either a discounting or a compounding process for a set of annual cash flows, \$100 per year, beginning one year from today. First click on “Present Value.” The illustration shows each annual cash flow discounted to present value at the rate r for periods of one to five years, according to when the cash flow is received. To begin, set the discount rate slider to $r = 5\%$. The illustration shows a present value of \$432.95. At lower discount rates, the present value will be higher—try 3% instead of 5% and observe the present value rising to \$457.97. At a rate of 7%, the present value falls to \$410.02.

More generally, the formula for the calculation of the present value (at $t = 0$) of a finite series of periodic cash flows is given by:

$$PV = \frac{Cf_1}{1+r} + \frac{Cf_2}{(1+r)^2} + \frac{Cf_3}{(1+r)^3} + \dots + \frac{Cf_T}{(1+r)^T}$$

$$PV = \sum_{t=1}^T \frac{Cf_t}{(1+r)^t}$$

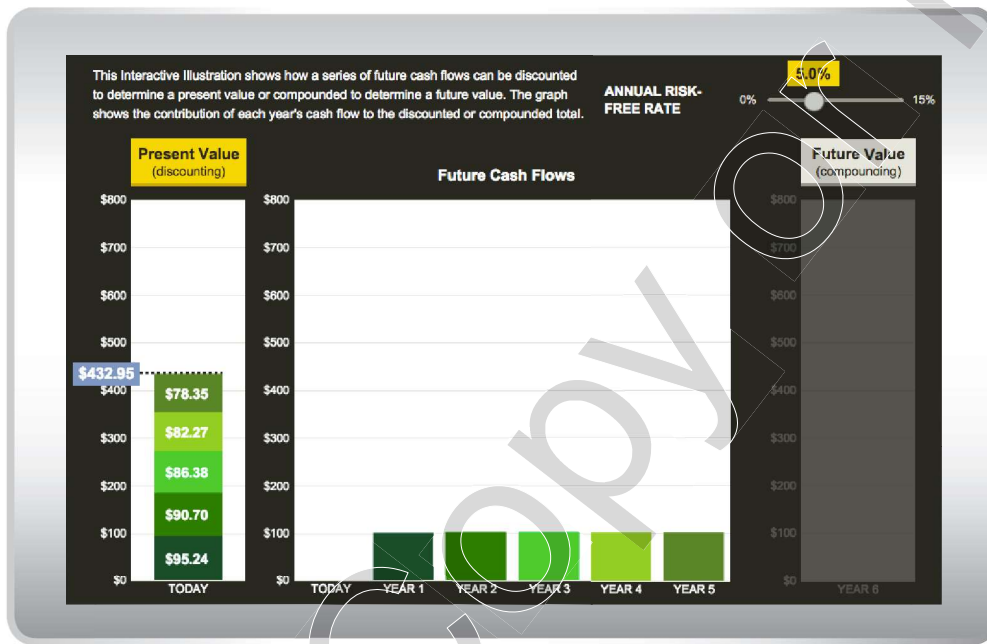


INTERACTIVE ILLUSTRATION 3

Present or Future Value of Multiple Cash Flows



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Now click on “Future Value.” The illustration computes the value of each \$100 cash flow at the end of year 6—one year after the last cash flow—by applying the appropriate compounding factors (or **FV factors**) and summing the result. Just as we saw in Table 1, the compounding factors are all greater than 1.0, and they increase as the investment horizon lengthens. As you toggle between present and future values in Interactive Illustration 3, holding the interest rate constant, observe that the future values are always greater than the present values.

The formulas we have developed for discounting and compounding multiple cash flows over multiple periods are quite general. The cash flows may be uneven and/or irregular, and the formulas can easily accommodate this. That is, within the stream of cash flows $Cf_1, Cf_2, Cf_3, \dots, Cf_T$, all cash flows may be the same or all may be different, and any may be missing (that is, equal to zero). Further, note that the length of a period may be longer or shorter—a year, six months, one month, and so forth—as long as the associated discount rate matches the period (i.e., annual interest rates for yearly discounting, semiannual rates for semiannual discounting, and so forth). We next turn to the problem of adjusting the length of the compounding or discounting period.

2.4 The Frequency of Compounding and Discounting

So far, all of our examples have utilized annual compounding or discounting. What if we performed the compounding or discounting more (or less) frequently?

Consider again investing \$1,000 for one year—say, in a guaranteed bank deposit—at a rate of 5%. But suppose that instead of paying 5% interest at the end of the year, the bank will pay interest semiannually. Specifically, the bank agrees to pay 2.5% interest every six months. This is beneficial, because we will now begin to receive interest on interest after six months instead of having to wait a whole year. At the end of one year, our deposit is worth $\$1,000 \cdot 1.025 \cdot 1.025$ or $\$1,000 \cdot (1.025)^2 = \$1,050.63$. The **effective interest rate** is 5.063% (i.e., $(1.025)^2 - 1 = 5.063\%$) rather than the quoted 5.00% (the 5% rate is often referred to as a **simple interest rate**). If the bank offered monthly compounding, we would receive $0.05/12 = 0.4167\%$ per month for an annual total of

$$\$1,000 \cdot (1.004167)^{12} = \$1,000 \cdot 1.05117 = \$1,051.17$$

Compounding daily, we would receive

$$\$1,000 \cdot \left(1 + \frac{0.05}{365}\right)^{365} = \$1,051.27.$$

For compounding periods less than one year, the formula for the effective annual interest rate is

$$\text{Effective interest rate} = \left(1 + \frac{r}{m}\right)^m - 1$$

where r is the simple annual rate, and m is the number of compounding periods per year.

As m approaches infinity, the length of each compounding period approaches zero, and the value of $\left(1 + r/m\right)^m$ approaches e^r , where e is the base of the natural logarithm, approximately 2.7183.^c

When the compounding interval converges to zero, we say that we are compounding *continuously*. As always, there is an analogous expression for *discounting* continuously. One

^cThe base of the natural logarithm, e , is an important natural constant, sometimes referred to as Euler's number. It was discovered by Jacob Bernoulli in the course of his work on compound interest. It can be defined in many ways, perhaps most compactly as the sum of the infinite series:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

dollar received at the end of a year and discounted continuously at the annual rate r has a present value equal to e^{-r} .

These simple expressions are easily adjusted to reflect continuous compounding and discounting over longer periods. The future value of \$1 compounded continuously for T years at an annual interest rate r is given by $FV = e^{rT}$. Suppose $r = 5\%$ per year and we want to compound continuously for 10 years. Then e^{rT} equals 1.649 (i.e., $e^{0.05 \times 10} = 1.649$) and the future value of \$1,000 compounded continuously at 5% for 10 years is $\$1,000 \cdot 1.649 = \$1,649$.

The present value of \$1 discounted continuously for T years at the annual rate r is given by $PV = e^{-rT}$. Accordingly, the continuous discount factor for 5% over 10 years is 0.60653 ($= e^{-0.05 \times 10}$) and the PV of \$1,000 to be received at the end of year 10 is $\$1,000 \cdot 0.60653 = \606.53 .

Why is this useful? Some investments actually are compounded or discounted continuously. More broadly, continuous compounding/discounting is often a convenient approximation, because it simplifies mathematics without significantly distorting the economics of the problem at hand. Perhaps the best example of such use is in option pricing models, some of which are much simpler mathematically because key variables are compounded or discounted continuously rather than discretely over time.

Exhibit 5 shows effective annual interest rates at different compounding frequencies, but now applied to a higher interest rate; $r = 10\%$ instead of 5% (the higher rate of 10% magnifies the compounding factor to highlight differences across interval lengths). As you would expect, the shorter the compounding interval, the more closely discrete compounding approximates continuous compounding. In Exhibit 5 the effect of more frequent compounding is greatest when the compounding period, m , changes from annual to semiannual. Beyond that, the effective annual rate increases as m increases, but at a diminishing rate. As the frequency goes from daily to continuous compounding, the difference in the effective annual rate is very small indeed, even at the higher annual rate of 10%.

EXHIBIT 5

The Future Value of \$1 Million in One Year at Various Compounding Frequencies

Simple Interest Rate r	Compounding Frequency	Number of Compounding Periods/Year m	$\left(1 + \frac{r}{m}\right)^m$	Effective Annual Interest Rate	Value of \$1,000,000 at $T = 1$
10%	Annual	1	1.100000	10.0000%	1,100,000
10%	Semiannual	2	1.102500	10.2500%	1,102,500
10%	Quarterly	4	1.103813	10.3813%	1,103,813
10%	Monthly	12	1.104713	10.4713%	1,104,713
10%	Daily	365	1.105156	10.5156%	1,105,156
10%	Continuous	∞	1.105171	10.5171%	1,105,171

Interactive Illustration 4 lets you observe the discounted or compounded values of \$100 at earlier or later points in time at varying interest rates. Here you can change both the discount rate and the compounding interval to observe present and future values at different compounding or discounting frequencies. When does the length of the compounding period have the greatest effect on future value—at lower or higher interest rates?

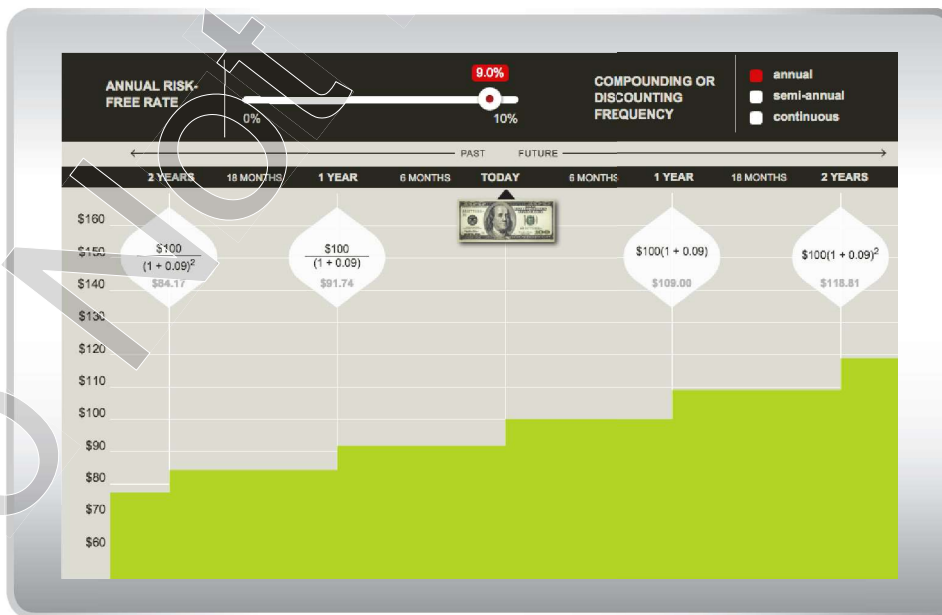


INTERACTIVE ILLUSTRATION 4

Time Value of \$100 with Compounding Frequencies



Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2pKuV72



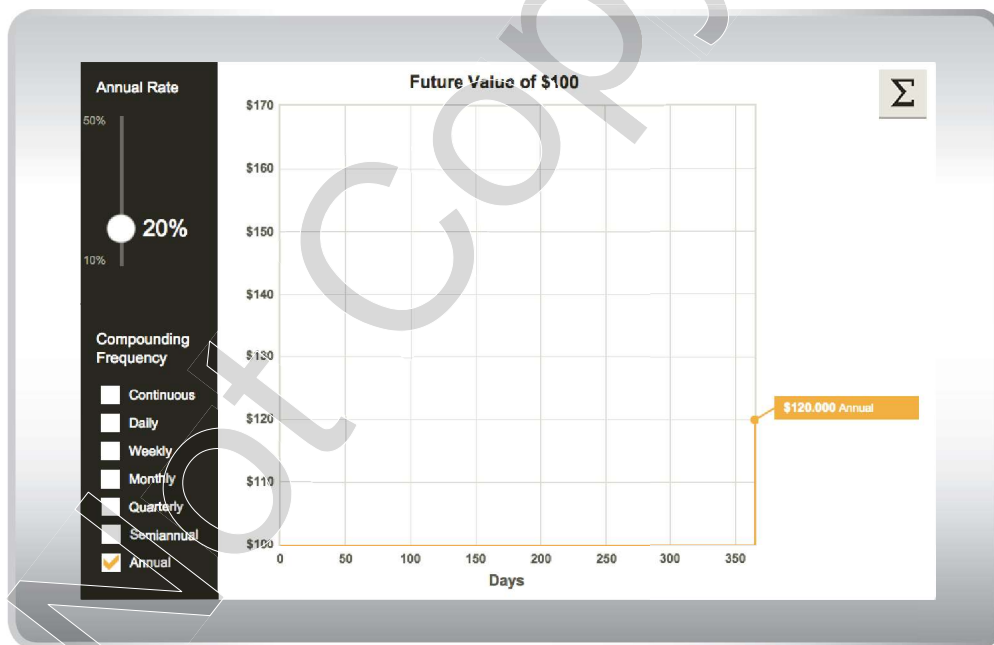
Interactive Illustration 5 lets you compare compounding intervals for a hypothetical bank balance of \$100 earning a specified simple annual interest rate for a single period. Begin with the annual interest rate slider set at 20% per year and check the “Annual” box for compounding frequency. The graph shows a future value of \$120 after one year (and no interim payments of interest). Now click “Semiannual” compounding frequency (without deselecting “Annual”). You can observe an interim interest payment of \$10 halfway through the year, which results in a balance of \$110 and interest on interest for the next six months, resulting in an ending value of \$121. Click on each of the compounding frequency boxes in order and observe the corresponding set of interim interest payments and the resulting higher year-end value. Finally, try out different interest rates to see how much more significant the compounding frequency is at higher rates.



INTERACTIVE ILLUSTRATION 5 Effect of Compounding Frequency



Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2DXDPmd



Equivalent Interest Rate Conversions

We have just seen how a simple annual interest rate implies a higher effective annual interest rate when the compounding frequency increases. For example, a simple annual rate of 5% implies an effective annual rate of 5.063% when compounded semiannually. Given a simple annual rate and a compounding frequency, we can always calculate a corresponding effective annual rate using the formula in the text.

However another, different, calculation is also common. We can convert a simple annual interest rate to an **equivalent interest rate** (for example, a semiannual rate) such that the future value (or present value if we are discounting) remains unchanged *despite* the different compounding (discounting) interval. Put another way, we can set the simple annual rate equal to the effective annual rate, and then solve for the implied (“equivalent”) semiannual rate.

To be concrete, let’s return to the example of a bank that pays simple interest of 5.0% annually on a deposit of \$1,000. Suppose we ask the bank to pay us semiannually instead of annually (we are hoping to receive 2.5% every six months, which we know is more valuable than 5.0% paid at the end of the year). But the bank understands our hope and instead offers the following proposition: you, the customer, may name any compounding frequency you like and we, the bank, will tell you what rate we offer at that frequency.

The bank can convert its simple 5.0% annual rate to an equivalent rate for any compounding frequency we may name. If we request semiannual compounding, the bank will offer a lower interest rate—specifically, 2.4695% per six months instead of 2.5%. Then at the end of one year our \$1,000 deposit will be worth $\$1,000 \cdot (1.024695)^2 = \$1,050.00$, exactly the same amount as if we had been paid 5.0% simple interest. The bank can calculate an equivalent semiannual rate that gives exactly the same future value for our deposit as its simple annual rate. In other words, the bank can choose a semiannual rate such that the simple rate equals the effective rate. The equivalent semiannual rate is given by:

$$\text{equivalent semiannual rate} = (1+r)^{\frac{1}{2}} - 1$$

where r equals the simple annual rate.

More generally, for higher frequency compounding, the equivalent rate is given by:

$$\text{equivalent rate} = (1+r)^{\frac{1}{m}} - 1$$

where r is again the simple annual rate and m is the number of compounding periods per year. So if we asked for daily compounding, the bank’s equivalent daily rate will be:

$$\text{equivalent daily rate} = (1+r)^{\frac{1}{365}} - 1 = (1.05)^{\frac{1}{365}} - 1 = 0.01337\%$$

In each case, compounding the equivalent rate at the specified frequency will result in the simple annual rate of 5%:

$$\text{semiannual: } (1.024695)^2 - 1 = 5\%$$

$$\text{daily: } (1.0001337)^{365} - 1 = 5\%$$

To summarize, *equivalent* rates for higher frequency compounding are *lower* than the simple annual rates. In contrast, *effective* annual rates are *higher* than the simple annual rates whenever the compounding frequency is greater than once per year.

It may be tempting to ask at this point: Which rate is right? Which should we use? It depends on what we're trying to figure out and on how a given rate has been quoted. In the bank deposit example, it is essential to know how much interest the bank will actually pay: Is it going to pay half of 5% semiannually, or will it pay 2.4695%? For some financial instruments, such as corporate bonds or home mortgages, long-standing conventions and/or regulations may dictate the basis on which interest rates are quoted or disclosed. For others, there is no substitute for reading the contract carefully and working out the numbers yourself.

Performing calculations for present and future values by hand or with a handheld calculator becomes cumbersome for more than a few time periods, but the necessary formulas are easily programmed in Excel or are accessible as Excel functions. The convenience of Excel's preprogrammed functions have made them ubiquitous in many real-world financial analyses, but using them properly and avoiding errors requires an understanding of the concepts that underlie them. The concepts are the subject of this reading; see the Supplemental Reading for notes on compounding and discounting using Excel.

Additional Excel Functions

In addition to the functions discussed above, a few other Excel functions perform calculations related to the discounting and compounding problems covered in the reading.

FV SCHEDULE calculates the future value of an initial principal amount after applying a series of (potentially unequal) *compound* interest rates over equal periods. It is therefore useful if interest rates are expected to change over the investment horizon.

IPMT and **PPMT** are used to determine the interest and principal payments, respectively, for an amortizing loan over a given period. That is, these functions divide a constant periodic payment into components of interest and principal on the basis of the given interest rate.

NOMINAL returns the annual nominal interest rate for a loan (also known as the APR, or annual percentage rate) given the *effective* rate and compounding frequency. Conversely, the **EFFECT** function returns the effective interest rate given the nominal rate and compounding frequency. For example, a loan with a nominal rate (APR) of 10% and monthly compounding has an effective interest rate of 10.47%.

4 KEY TERMS

amortizing loan A type of loan in which the principal is repaid over the life of the loan, via periodic payments containing both interest and repayment of a portion of the outstanding principal balance.

annual percentage rate (APR) The simple rate of interest on a loan (before compounding). Also referred to as the *nominal rate of return*.

annuity A stream of constant periodic payments paid or received every period for a finite number of periods.

annuity factor (PV factor) The mathematical factor applied to the periodic cash flow of an annuity to determine its present value:

$$\left(\frac{1}{r} - \frac{1}{r(1+r)^T} \right)$$

compounding The process of reinvesting the return earned on an investment, which causes value to accumulate, or grow, over time.

compounding factor (FV factor) The mathematical factor applied to determine the future value of a single amount T years in the future: $(1+r)^T$.

discount rate The rate at which future cash flows are discounted to determine their present value.

discounting The process of calculating the present value of an amount to be received in the future. The application of a discount rate in this calculation causes the present value to be less than the future value.

discount factor (PV factor) The mathematical factor applied to determine the present value of a single amount T years in the future: $1/(1+r)^T$. *See also* annuity factor.

effective interest rate The annual rate of interest implied when compounding takes place more than once per year, calculated as

$$\left(1 + \frac{r}{m}\right)^m - 1$$

where r is the simple annual rate and m is the number of compounding periods per year.

equivalent interest rate A rate corresponding to a given compounding/discounting frequency (say, semiannual) that has been expressly calculated to give the same FV or PV as a rate corresponding to a different frequency (say, annual), calculated as

$$(1+r)^{\frac{1}{m}} - 1$$

That is, one may speak of “converting” an annual rate to an equivalent rate on a semiannual, monthly, or weekly basis.

future value The value at a specific point in the future of a stipulated amount of cash today. Future value includes the stipulated amount as well as interest earned and compounded over the appropriate period.

growing (shrinking) perpetuity A perpetuity in which the periodic payment increases (or decreases) by a constant rate of growth (or decay).

inflation rate The rate at which the general level of prices for goods and services, measured in a certain currency, is rising. Implicitly, this reflects the rate at which the currency’s purchasing power is falling.

mortgage A collateralized loan, typically secured by real estate, which also exhibits the characteristics of an *annuity* (periodic payments of an equal amount) and an *amortizing loan*.

nominal rate of return The quoted simple rate of interest paid (or received) on a loan (or deposit). When the nominal rate is an annual rate, it is also sometimes called the *annual percentage rate* or APR. (It also can be expressed as a periodic rate by dividing the annual rate, r , by a number of periods per year, m —i.e., r/m .) Nominal rates do not reflect compounding (*see also* simple interest rate). In different usage, the term *nominal* may also be used to reflect a contrast with a “real” (i.e., inflation-adjusted) rate of return.

perpetuity Usually a stream of periodic, constant cash flows paid or received forever. *See also* growing (shrinking) perpetuity.

present value The value today of a cash flow to be paid or received in the future; the discounted value of a series of future cash flows.

principal The amount lent to a borrower, to be repaid at a future date or dates, and on which the borrower pays a stated rate of interest.

real rate of return The inflation-adjusted rate of return, approximated by the nominal rate of return less the rate of inflation.

risk-free interest rate The rate of interest earned on a security for which promised cash flows are certain to be paid/received.

simple interest rate An interest rate for which the compounding interval and the payment period are the same (most commonly annual). For example, a simple annual interest rate of 5% denotes 5% interest per year, compounded annually.

time value of money The return investors can earn over a period of time without taking any risk of loss of their money.

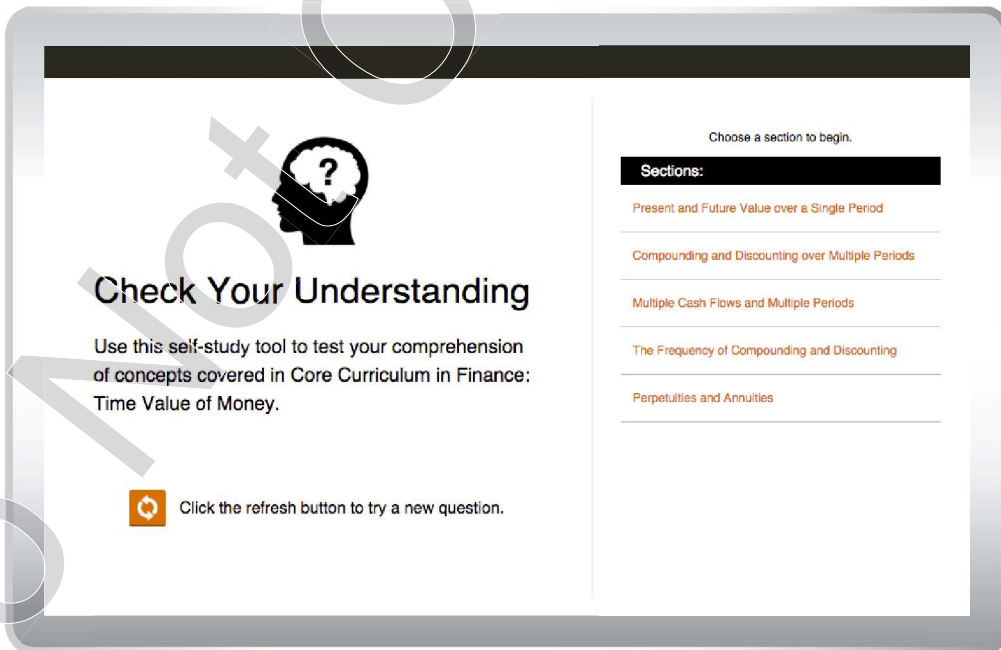
5 NOTATION

Cf	Cash flows
FV	Future Value
g	Growth rate
m	Number of compounding periods per year
PV	Present Value
r	Discounting or compounding rate
t	Date; time period
T	Number of time periods (date of the endpoint)

6 PRACTICE QUESTIONS



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